ICT2217

Network Security

Lab 3.6 Notes:

Cisco Flexible NetFlow

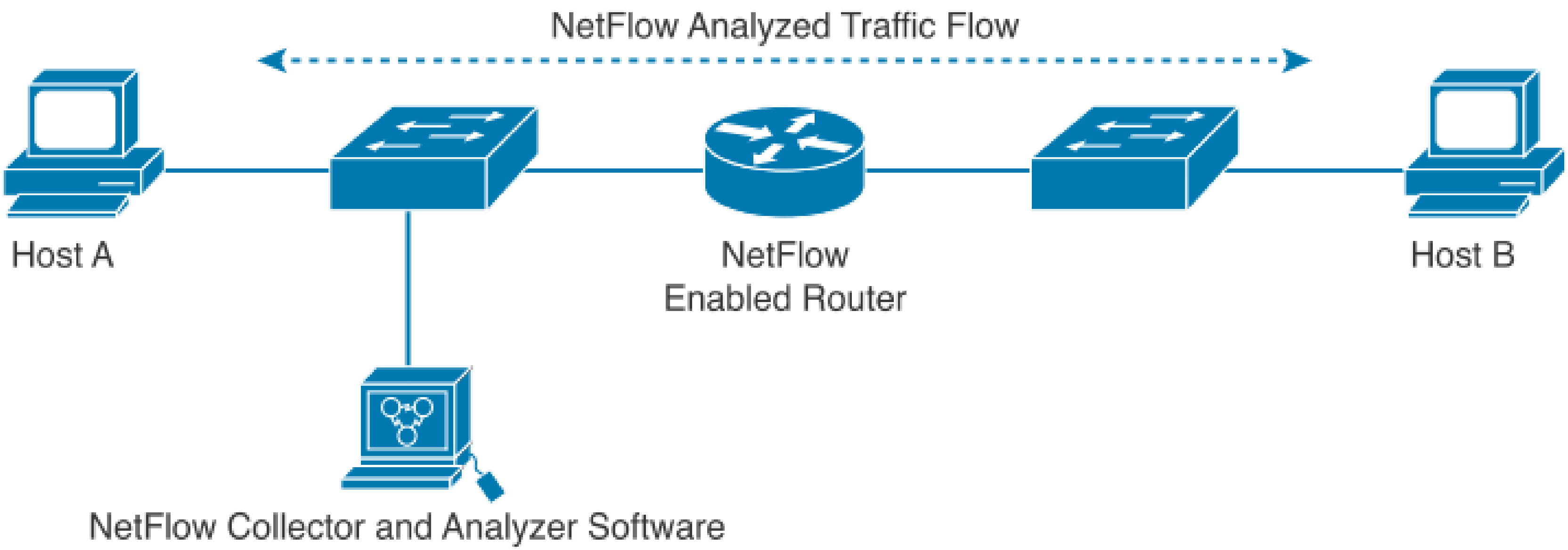
Lab Exercise

1

Configure Flexible NetFlow to provide statistical records of different IP packets that flow through a router.

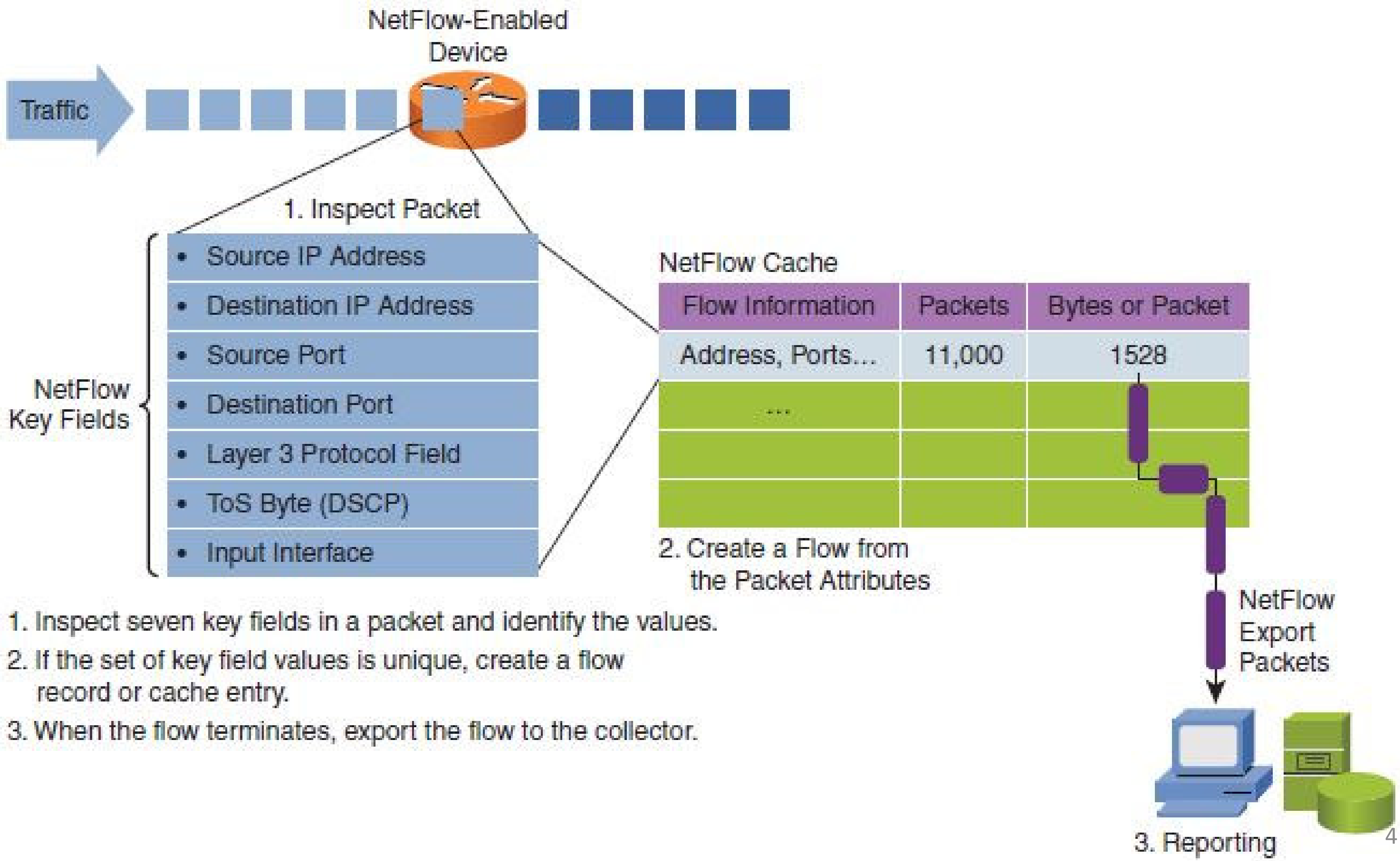
To support network traffic monitoring, Cisco has developed NetFlow to collect statistics on IP packets flowing through network devices.

When enabled, NetFlow keeps statistics of different IP packets flowing through a router in a NetFlow cache, which can then be exported to a NetFlow Collector for centralized logging and analysis.



NetFlow has proven valuable and has evolved to variants like Flexible Netflow (RFC 3954), IPFIX (RFC 7011), sFlow (RFC 3176) and JFlow (Juniper).

A flow is defined as a unidirectional sequence of packets that pass through a network device having the same values for certain key fields in the packet headers, examples:



Today, NetFlow version 9 is supported by Cisco allowing flexible configuration of matching rules and collection statistics, hence also called flexible NetFlow.

(1) Example of configuring matching rules and collection statistics using flow record:

```
R1(config)# flow record my-record
R1(config-flow-record)# match ipv4 source address
R1(config-flow-record)# match ipv4 destination address
R1(config-flow-record)# match transport source-port
R1(config-flow-record)# match transport destination-port
R1(config-flow-record)# match ipv4 protocol
R1(config-flow-record)# match ipv4 tos
R1(config-flow-record)# match interface input
R1(config-flow-record)# collect counter bytes
R1(config-flow-record)# collect counter packets
```

Apply specific matching rules and collection statistics configured onto the interface of the router as follows:

(2) Configuring flow exporter:

```
R1(config)# flow exporter my-exporter
R1(config-flow-exporter)# destination 10.1.10.100
R1(config-flow-exporter)# transport udp 99
R1(config-flow-exporter)# source lo0
R1(config-flow-exporter)# export-protocol netflow-v9
```

(3) Configuring flow monitor:

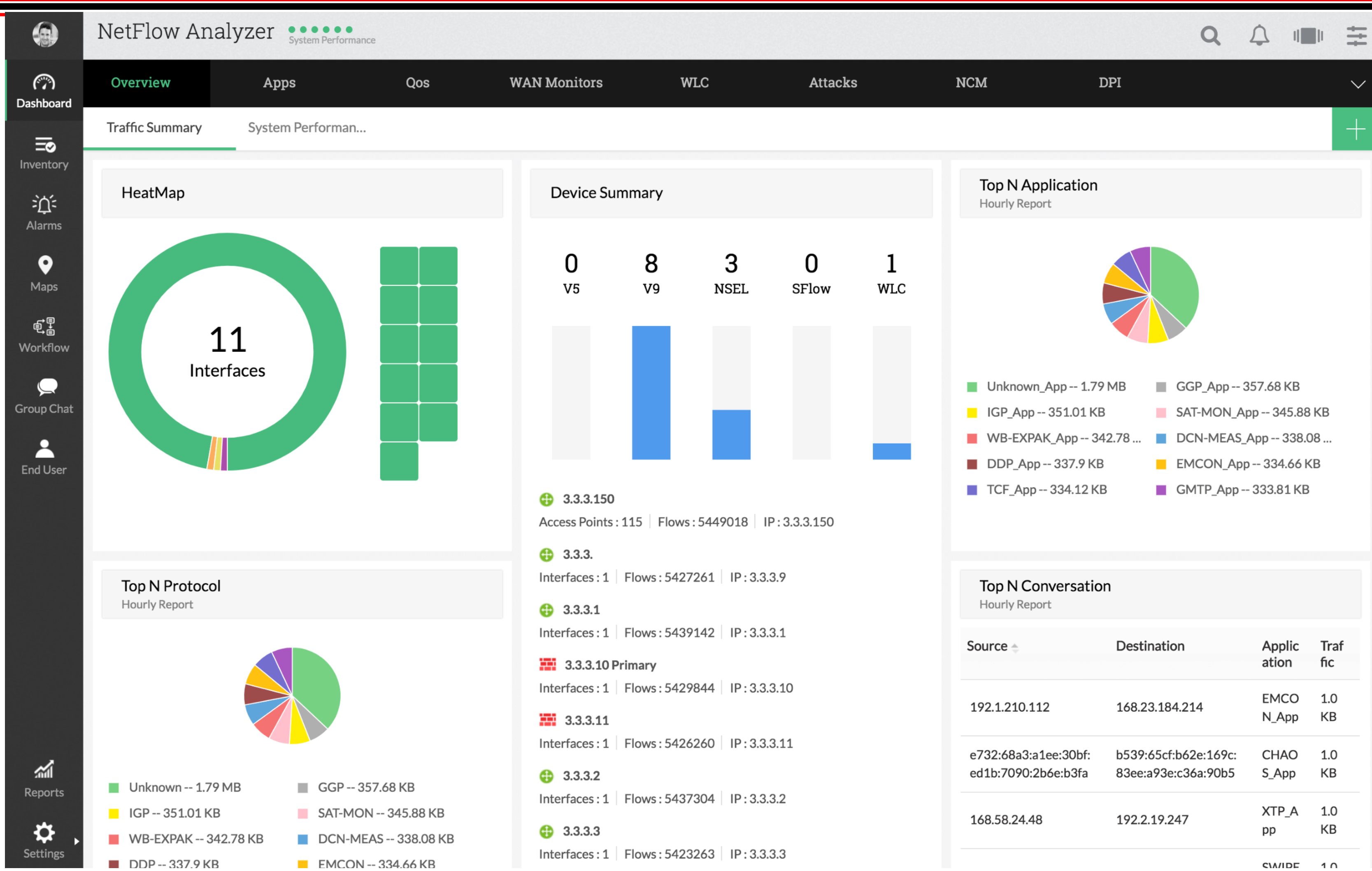
```
R1(config)# flow monitor my-monitor
R1(config-flow-monitor)# exporter my-exporter
R1(config-flow-monitor)# record my-record
```

(4) Applying flow monitor onto router interface:

```
R1(config)# interface g0/0
R1(config-if)# ip flow monitor my-monitor {input|output}
```

An example of NetFlow collector is the NetFlow Analyzer by ManageEngine which you can try in your team project.

https://www.manageengine.com/products/netflow/download-free.html



Finally, instead of separate logs, SIEM (Security Information and Event Management) solution is getting common to provide unified real-time correlation of all logs and threat alerting.